

Preliminary Economic Feasibility Study to Establish a Light Industrial Zone in the Municipality of Zarqa

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Executive Summary

The idea of the proposed Project is to establish a Light Industrial Zone away from the commercial and residential center of Zarqa. The land is planned to be a 500 donums Light Industrial Zone that will attract as much as possible of craftsmen (i.e. especially construction industries such as concrete bricks plants and stone and marble processing plants) and entrepreneurs, to reduce the problems associated with their concentration in the city and the accompanied pollution and noise. Therefore, the city will include a number of units (spaces) required by craftsmen, and the possibility of renting them to owners of commercial stores and restaurants.

Zarqa has multiple distributed industrial zones around the city, scattered across the governorate. The proposed Project is unique in Zarqa Municipality; consequently it will attract merchants and craftsmen who are willing to improve their businesses. Furthermore, the proposed Project will aim to mitigate the noise caused by concentration of crafts centers in the middle of the city. The proposed Project will offer suitable spaces for craftsmen to facilitate their day-to-day works and constant uploading and offloading, which create a burden on them in crowded residential areas. On the other hand, the Project will contribute significantly to customer service and provide craft service of all types in one place, which will facilitate the process of searching for their needs. Therefore, the essential advantage for customers is existence of all types of crafts they need in one place. For example, the need for brick plant for construction purposes, accompanied by the need for blacksmithing to meet the requirements of construction. Presence of these services in one place makes it easy for buyers and save their time and efforts in moving between locations of craftsmen, which are scattered randomly as it is now.

Major Results of the Study

- ✓ Investment cost of this Project is estimated at JD 3,774,680. This amount includes cost of land, machinery and equipment, vehicles and means of transport, furniture and fixtures, pre-operating expense and initial capital. (30%) will be financed by loans amounting JD 1,132,404 and the remaining (70%) of the Project will be financed through self-financing (owners' equity) amounting JD 2,642,276. The loan ratio is calculated according to the policy of the Investment Bank, so the bank financing a maximum amount of 30% of the Project's value for real estate projects.
- ✓ The expected statement of income is based on subtracting expenses from revenue, where the retained earnings of the Project are positive from the first year of the Project.
- ✓ The cash flows of the Project come from operational, investment and financing activities, where cash flows at the end of the year are the aggregate of all cash inflows and outflows from all activities.
- ✓ Net current value of cash flows of the Project is negative (844, 097). The team the Professional Group calculated the internal rate of return (IRR) on investment, aiming to verify feasibility of the Project economically, where the IRR rate was 9.11%, which is lower than the Weighted Average Cost of Capital (WACC) 12.97%.

✓ All financial indicators indicate that the Project is profitable and economically feasible, due to the large volume of investment. For the purpose of examining profitability and feasibility of the Project, a sensitivity analysis for the Project was developed in order to determine the most influential factors that have impact on the results and financial ratios of the Project. The following table shows the results of sensitivity analysis:

Table 1: Sensitivity Analysis

		For Project			For Owners' Equity		
	Item	Internal rate of return	Net current value	Capital recovery period	Internal rate of return	Net current value	Capital recovery period
	Current Position	9.11%	(844,097)	More than 10 years	9.75%	(589,229)	More than 10 years
Capital costs							
1	10% decrease in capital costs	10.67%	(468,872)	More than 10 years	11.56%	(241,657)	More than 10 years
2	20% decrease in capital costs	12.48%	(93,648)	9	13.65%	105,916	9
3	25% decrease in capital costs	13.50%	93,964	8	14.83%	279,702	9
Operating revenue							
4	10% increase in revenue	10.93%	(465,597)	More than 10 years	11.87%	(210,709)	More than 10 years
5	20% increase in revenue	12.60%	(87,097)	9	13.82%	167,811	9
6	25% increase in revenue	13.40%	102,153	8	14.75%	357,071	9
Operating expenses							
7	20% decrease in operating expenses	10.30%	(600,091)	More than 10 years	11.13%	(345,308)	More than 10 years
8	30% decrease in operating expenses	10.87%	(478,089)	More than 10 years	11.80%	(223,348)	More than 10 years
9	50% decrease in operating expenses	11.96%	(234,083)	9	13.08%	20,573	More than 10 years
10	70% decrease in operating expenses	13.02%	9,922	9	14.31%	264,494	9
Indirect salaries							
11	20% decrease in indirect salaries of employees	9.39%	(787,700)	More than 10 years	10.08%	(532,876)	More than 10 years
12	40% decrease in indirect salaries of employees	9.66%	(731,304)	More than 10 years	10.40%	(476,523)	More than 10 years
13	60% decrease in indirect salaries of employees	9.93%	(674,907)	More than 10 years	10.71%	(420,169)	More than 10 years
Administrative expenses							
14	20% decrease in administrative expenses	9.15%	(835,506)	More than 10 years	9.80%	(580,645)	More than 10 years
15	40% decrease in administrative expenses	9.19%	(826,916)	More than 10 years	9.85%	(572,061)	More than 10 years
16	60% decrease in administrative expenses	9.24%	(818,325)	More than 10 years	9.90%	(563,478)	More than 10 years

1 Chapter I: Jordan Overview

1.1 Geographic and Demographic Reality



Area¹: 89,318 km²

Population: 9.9 Million (2017)

Population Growth Rate (2017): 7.9%

Key Economic Indicators (2017)²:

- GDP: USD 38.74 Billion
- Inflation Rate: 3.2%
- Unemployment Rate: 15.8%

Jordan sits in the heart of the Middle East region, bordering by Saudi Arabia, Palestine, Iraq, and Syria from the south, west, east and north, respectively. Jordan has access to the Red Sea through Aqaba city, which located on the northern tip of the Gulf of Aqaba. The following table shows the geographic and demographic reality in Jordan⁴:

Table 2: View of the Geographic and Demographic Reality in Jordan

	Description
Location	Middle East, bordering by Saudi Arabia, Palestine, Iraq, and Syria from the south, west, east and north, respectively.
Geographic Coordinates	31° North and 36° East
Area	Total Area: 89,318 km ² , Land Area: 88,778 km ² , Water Area: 540 km ²
Coast line	26 km
Climate	The weather is hot and dry in the summer and nice, humid and rainy in winter (runs from November to April)
Lower and higher heights	Lower height: The Deas Sea, falling 408 m, below sea level. Higher height: Umm Al-Dami summit, with estimated height of 1854 m above sea level
Natural Resources	Phosphate, Potash and Oil Shale

¹Department of Statistics: Report of main results on general census of population and housing 2017 - Access date (12.10.2017).

²World Fact Book (CIA FACTBOOK) <https://www.cia.gov/library/publications/the-world-factbook/geos/jo.html> - Access date (12.10.2017)

³World Fact Book (CIA FACTBOOK) <https://tradingeconomics.com/jordan/inflation-cpi> - Access date 12.10.2017)

⁴World Fact Book (CIA FACTBOOK) <https://www.cia.gov/library/publications/the-world-factbook/geos/jo.html> - Access date (12.10.2017)

Jordan is divided in to 12 governorates (The capital, Balqa, Zarqa, Madaba, Irbid, Mafrq, Jerash, Ajloun, Karak, Tafila, Ma'an, and Aqaba). According to the statistics issued by Department of Statistics for 2016, the population of Jordan reached about 9,798,000. The following table shows the population, area and population density⁵ of each governorate as follows⁵:

Table 3: Population, area and population density for each governorate in the Kingdom

Governorate	Population	Area (km ²) ⁷	Population Density
Amman	4,119,500	7,579	544
Balqa	505,400	1,119	452
Zarqa	1,403,000	14.7	295
Madaba	194,500	940	207
Irbid	1,819,600	1,572	1158
Mafrq	565,300	26,551	21
Jerash	243,700	410	594
Ajloun	181,000	420	431
Karak	325,500	3,495	93
Tafila	99,000	2,209	45
Ma'an	148,100	32,832	5
Aqaba	193,400	6,900	28
Total	9,798,000	88.88	110.4

⁵Central Bank of Jordan - http://www.cbj.gov.jo/uploads/result_ar.pdf - Access date (12.10.2017).

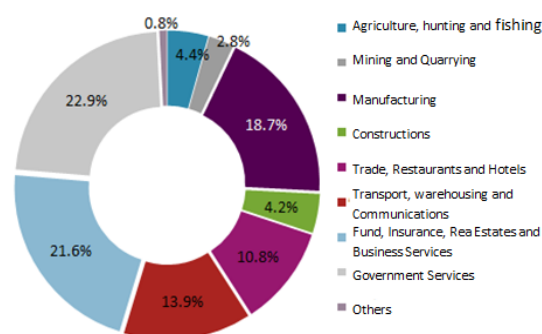


Figure 1: Gross Domestic Product by Economic Sectors

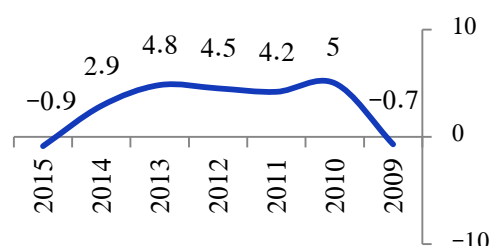


Figure 2: Inflation ratios in Jordan for the period (2009-2015)

The Gross Domestic Product (GDP) in Jordan consists of several sectors, such as agriculture sector, hunting, fishing, mining and quarrying, manufacturing, construction, trade, restaurants and hotels, transport, storage, communications, financial services, insurance, real estates and business services, government services and other sectors. The government services sector has the higher rate, which constitute 22.9% of the GDP. The following figure shows contribution of each sector to the GDP in Jordan for the year 2016⁶. Based on the preliminary estimated of the International Monetary Fund (IMF), the economic growth in Jordan will be accelerated in the coming years, so the growth rate of GDP will be 2.5% in 2015 and 4% in 2021⁷.

Inflation

Inflation is defined as a constant increase in price levels in a country, which lead to weakening the purchasing power in that country and high cost of living in general. The importance of knowing the change in inflation rates as an economic indicator in any country is considered as an indicator of economic stability in that country, or ability governments to control the economy through their financial and economic policies. Furthermore, the inflation rate in a country may describe the trade policy of that country and its openness.

According to the information issued by the Department of Statistics on inflation in Jordan, the average consumer prices for the first two months of 2017 decreased by 3.5% compared with the same period of 2016⁸, while the inflation rate reached 2.28% during the years (2013-2015)⁹. The major reasons beyond the changes in inflation rates are: The unemployment rate rises to 15.8% due structural disparities in the labor market and acquisition of low-paid expatriates, especially the Syrian labor, on great number of new job opportunities in the economy. In the light of unstable oil prices and associated services in the global markets.

⁶Department of Statistics - http://www.dos.gov.jo/dos_home_a/main/economic/nat_account/sel2/nat_6/1.pdf - Access date (12.10.1027).

⁷International Monetary Fund (2016) Database of Global Economic Expectations- <http://www.imf.org/external/pubs/ft/weo/2016/01/pdf/text.pdf> - Access date (12.10.2017).

⁸Department of Statistics - <http://web.dos.gov.jo/%D8%A7%D9%84%D8%B9%D8%B1%D8%A8%D9%8A%D8%A9-4-6-%D8%A7%D8%B1%D8%AA%D9%81%D8%A7%D8%B9-%D9%85%D8%B9%D8%AF%D9%84-%D8%A7%D9%84%D8%AA%D8%B6%D8%AE%D9%85-%D9%84%D8%B4%D9%87%D8%B1-%D8%B4%D8%A8%D8%A7%D8%B7/?lang=ar> - Access date (12.10.2017).

⁹Inflation rate for the last three years - Central Bank

Jordan is one of the leading countries in the region in economic openness experience. In spite of its limited resources, it has managed to improve its economic relationships with the Arab countries and worldwide through concluding a number of agreements, such as the Greater Arab Free Trade Area Agreement (GAFTA), which contributed in facilitating trade exchange between the member countries. Jordan also entered into a partnership agreement with the European Union and a free trade agreement with the United States, in addition to signing a number of agreements to protect and promote investment and prevent double taxation with a number of Arab countries.

According to information from the CIA-World Factbook, the value of Jordan's exports for 2016 was USD 7.124 billion¹⁰, while the value of imports was USD 17.86 billion¹¹. Jordan's main exports are clothing, fertilizers, potash, phosphate, vegetables and pharmaceuticals. The main imports are crude oil, refined petroleum products, machinery and equipment of transport, iron and grain. The following table shows a number of free trade agreements, which Jordan currently associated with, according to information from the Ministry of Industry and Trade¹²:

Table 4: Free trade agreements with Jordan

Agreement	Signing Date
Greater Arab Free Trade Area (GAFTA)	19/2/1997
Jordanian-European Partnership	24/11/1997
The United States-Jordan Free Trade Agreement	24/10/2000
Free trade Agreement with EFTA States	21/6/2001
Free Trade Area Between the Arab Mediterranean Countries (Agadir)	25/2/2004
Free Trade Area with Singapore	16/5/2005
Free Trade Agreement with Canada	28/6/2009
Free Trade Area Agreement with Turkey	1/12/2009

¹⁰CIA-World Factbook <https://www.cia.gov/library/publications/the-world-factbook/geos/jo.htm> - Access date (12.10.2017)

¹¹CIA-World Factbook <https://www.cia.gov/library/publications/the-world-factbook/geos/jo.html> - Access date (12.10.2017)

¹²Ministry of Industry and Trade - <http://www.mit.gov.jo/EchoBusV3.0/SystemAssets/Documents/AR/%D9%86%D9%87%D8%A7%D8%A6%D9%8A%20-%D8%A7%D9%84%D8%AA%D9%82%D8%B1%D9%8A%D8%B1%20%D8%A7%D9%84%D8%B3%D9%86%D9%88%D9%8A%202015.pdf> - Access date (12.10.2017).

1.2 Investment Environment

Jordan is characterized with its strategic location, since it is located at the confluence of three continents, which are Europe, Asia and Africa. Therefore, it is an ideal gateway to the Middle East, North Africa and beyond. Jordan provides a safe and convenient environment for businessmen and investment. It has an advanced legal system, a world-class infrastructure, an advanced communications system and easy access to markets through international agreements. The Jordanian human resources are also qualified, educated and highly competitive.

The Jordan Investment Commission (JIC) grants incentives to both Jordanians and non-Jordanians represented by exemption customs duties, exemptions from income tax and social services tax, in addition to unrestricted transfer of capital and profits. Based on the information that was obtained by the professional team from JIC, the following table shows the annual size of investments benefiting from the Investment Promotion Law for the years (2013-2015)¹³:

Table 5: Volume of investments benefited from the Investment Promotion Law for the years (2013-2015) distributed by type of investment and sector

Size of Investments	2013			2014			2015		
Type of Investments	Local Investments	Foreign Investments	Total Investments	Local Investments	Foreign Investments	Total Investments	Local Investments	Foreign Investments	Total Investments
Total (JD)	641,308,372	667,622,569	1,308,930,941	877,480,248	269,027,447	1,146,507,695	810,324,876	2,305,875,885	3,116,200,761

Table 6: Volume of investments benefited from the Investment Promotion Law for the years (2013-2015) distributed by sector

Sector	2013	2014	2015	Sector	2013	2014	2015
Industry	1,002,276,880	400,774,797	2,739,801,038	Means of Transport	82,000,000	62,188,875	126,044,000
Tourism	142,387,000	94,872,031	71,461,000	Recreational Cities	27,487,186	99,703,200	15,128,918
Agriculture	19,779,875	187,085,552	105,721,805	Others	1,000,000	25,350,000	9,420,000
Hospitals	34,000,000	270,533,240	48,624,000	Total (JD)	1,308,930,941	1,146,507,695	3,116,200,761

¹³ Jordan Investment Commission - 2016

There are many reasons that distinguish Jordan, in terms of attracting investments, including¹⁴:

1. Its unique and strategic location: As previously mentioned, it is located at the confluence of three continents, Europe, Asia and Africa. In addition to its location as a center point in the Middle East area, and its access to the Red Sea through Gulf of Aqaba, and other ports in neighboring countries.
2. Its stable political environment: represented by the wise leadership, which guarantees freedom of believe and opinion, and protection of private property.
3. Its free market economy: The concept of free economy is based on non-intervention by the government in economic activities and leaving the market self-regulated. This is manifested through the constant privatization of large organizations that owned by the government, and tangible progress in the legal and structural reforms.
4. Offering a bundle of incentives and exemptions, aiming to promote investment, including: exemption of projects in any of the sectors for 10 years, of income tax and social services tax by 25%, 50% or 75%, depending on location and sector of Project. In addition to presence of several incentives to encourage investment, which included in the Jordan Investment Promotion Law, under which JIC was established.
5. Its access to the major international markets: Trade agreements with neighboring countries and worldwide facilitate access of the Jordanian products to those countries.
6. Existence of many free zones and industrial cities: The Free Zones Corporation operates five public free zones, and about 30 private free zones. Additionally, there are six industrial cities distributed in the capital and the rest of governorates in the Kingdom.
7. Competent and qualified human resources: according to information provided Ministry of Labor in 2014¹⁵, the Jordanian labor force is estimated at 1.461 million. The labor force in Jordan is expected to increase as a result of the increased percentage of women joined the labor market. The human resources in Jordan are one of the most important resources. Jordan is exporting its labor force to neighbor countries, especially the Arab Gulf, for a long time, contributing significantly in providing a source of income to the country.
8. High level infrastructure, in addition to existence of an advanced communications sector in the Kingdom.

High quality of life: Investor enjoys a relatively low standard of living, high quality private and government education in Arabic, English and French, affordable health service according to the international standards, cultural centers, local and international business associations, developed intercommunicating community, traditional festivals, cultural entertainment and wealth of the archaeological sites, restaurants and entertainment clubs.

¹⁴ http://www.jordan.gov.jo/wps/portal/!ut/p/b1/04_Sj7Q0N7Y0MTU2NdeP0I_KSyzLTE8syczPS8wB8aPM4sMsvS3CvN0NDQwsLF0NPIP8nd1MPX2MHI1N9YMTi_RzoxwVAa9I-HM/

E-government – Access date (03.08.2016)

¹⁵ Annual report of Ministry of Labor 2014 - <http://mol.gov.jo/Documents/المكتبية/التقارير/2014/Ministry%20Of%20Labour%202014.pdf> - Access date (03.08.2016)

1.3 Ease of practicing business

According to information provided by the World Bank, practicing business in Jordan held the following scores on indicators related to business, compared with 190 countries. As shown in the following table, the scores obtained by Jordan in 2018 are better than in 2017 for most of aspects evaluated, as Jordan is ranked 103 on the ease of practicing business indicator.

Table 7: Rank of Jordan Economy for 2017-2018

Ease of Regulating Business	Rank of Jordan Economy in 2017	Rank of Jordan Economy in 2018	Change in Rank
Starting a business	106	105	1
Obtaining construction licenses	109	110	-1
Access to electricity	48	40	8
Ownership registration	96	72	24
Access to credit facilities	185	159	26
Protecting investors minority	165	146	19
Payment of taxes	79	97	-18
Trade in transit	50	53	-3
Enforcement of contracts	124	118	6
Settlement of insolvency	142	146	-4

1.4 Overview - The Municipality of Zarqa

1.4.1 Location

The Municipality of Zarqa is located at the northern-eastern part of Jordan, about 20 km east of the capital of Jordan, and embodies the largest governorate in the Kingdom. The area includes a large military base, as well as a Palestinian refugees' camp, which was founded in 1949 and has a population of 15,000 inhabitants. The governorate consists of three provinces, four districts and seven municipalities: Zarqa, Russaifa, Al Hashimiyeh, Al Dhulail, Al Hallabat, Al Azraq and Berain. Zarqa also enjoys a strategic location that embodies a link between the main roads, linking not only Zarqa to Amman, but also to Balqa, Jerash, Irbid and Mafrq. As for the climate, the area is characterized by a relatively sunny and moderate in the spring and autumn, a hot summer and cold winter.

1.4.2 Demographics

The population of Zarqa Governorate is 1,364,879. The population of the city of Zarqa is 648,882, where the numbers of males is 339,618 and females 309,264 as per a census conducted in 2015. 52.3% of the total population is male, while 47.7% female. The average size of the household in Zarqa is 4.9 persons, compared to 5.2 persons at the national level. The dependency ratio is 68%, with four persons dependent on one worker. Zarqa is a young community, with more than half of the population aged between 15 to 65 years old. The following table shows distribution of population in the governorate by administrative divisions:

Administrative Division	Males	Females	Total	Households
Zarqa City Province	421,672	380,593	802,265	168,781
Zarqa County	339,618	309,264	648,882	138,249
Berain	13,223	11,460	24,683	5,037
Al Dhulail	41,694	35,009	76,703	14,336
Al Azraq	27,137	24,860	51,997	11,159
Russeifa	257,888	224,012	481,900	94,674
Al Hashimiyeh	42,041	38,672	80,713	16,531
Zarqa Governorate	721,602	643,277	1,364,879	279,986

1.4.3 Average Household Income

According to the latest reports issued by the Department of Statistics in 2013, the average household income in Zarqa Governorate is JD 7,058.4 per family annually, which is less than the national average of JD 9,258 per family annually. The inhabitants spend most of their income on food, housing and transportation (35.1% for food, 27.5% for housing, 17.6% for transportation, and the rest for education, health care, cigarettes, etc.). In 2010, Zarqa recorded a poverty rate of 14.1% compared to 14.4% at the national level. The unemployment rate is 11.3% in the governorate, and may be slightly lower in the city, compared with 12% at the national level.

1.4.4 Public Services

Transportation: In general, the road network covers most of the municipal areas and provides transportation services to local inhabitants, and links Zarqa to nearby cities. There are two bus and taxi stations and domestic complex for light cars. Queen Alia International Airport is the nearest airport to the Municipality, which is about 40 minutes away. There is no railway or train station in the area.

Health Care: There are comprehensive health care facilities in the city, including eight hospitals with 940 beds: Zarqa Governmental Hospital and Prince Faisal Ben A-Hussein Hospital, in addition to five private hospitals with 315 beds, 44 health care centers, 36 maternity and childhood centers, 30 dental clinics, 206 pharmacies and one infectious disease center. Zarqa suffers from a lack of specialized medical equipment and qualified medical staff.

Public Security: There are 10 police stations and 15 fire stations, all under administration of a main directorate.

Solid Waste: 100% of the solid waste is collected daily, through five rounds at different locations to collect 400 to 500 tons of wastes, which includes 60% organic waste and the rest are distributed on plastic materials, glass, paper, cardboards and metals. There is no landfill within the boundaries of the Municipality, but there is a diversion station with an area of 100 donums, about 7 km away from the city, where quantities of waste collected from the city are put up.

Sewage Treatment: about 72% of houses in Zarqa Governorate depend on a sewage network, while 28% depend on cesspits.

Water: The water network covers more than 70% of the Municipality's areas. The Municipality of Zarqa suffers from many concerns related to water supply, scarce water resources and low level of artesian wells.

Energy: The electricity network covers 99% of the areas, and a broadband internet is available.

1.4.5 Educational Opportunities

The number of schools in Zarqa is 246 schools, 31% of them are rented or private schools. Number of students is 98,801 male and female students, number of students of the First Education Directorate of Zarqa is 98,811 students and number of students in the governorate is 194,307 students. Number of schoolteachers is 9,228. The success rate in the general secondary exam (Tawjihi) is one of the highest in the Kingdom. There are a number of distinguished universities and colleges in Zarqa: The Hashemite University, Zarqa Private University and Al Balqa Applied University, which offer many educational opportunities in the fields of art, science, research, education, nursing and engineering. Zarqa also has vocational training institutes.

1.4.6 Cultural and Recreational Opportunities

There are many cultural centers and libraries in the Municipality of Zarqa, in addition to many sports centers and youth clubs, noting that the Municipality includes a cultural directorate.

1.4.7 Economic Activities and Jobs

The Municipality of Zarqa is a major industrial area and an active commercial sector. There are 408 large industrial enterprises in the governorate: tanning factories, medical and treatment industries, engineering and IT, plastics, rubber, paper and cardboard. The city also has large commercial areas, as well as 11,000 retail stores, offices and commercial establishments.

1.4.8 Governance

Zarqa currently has an elected council with a budget of JD 24,312,544 IN 2014. 75% of the budget was spent on salaries and benefits of employees. Number of employees in the Municipality is 4,225 employees, according to the statistics of the HR Department, 6.2% of them have the general secondary certificate, 77.61 below the high school 7.85% have diplomas while 6.43% have BA/BSc degrees and 0.2% of them have master's degree.

The Investment Unit realized revenue of JD 2,282,432 and 4,020 jobs. The projects included a recycling plant, commercial warehouses, shopping malls, car parking lots, kiosks, billboards and recreational facilities. The Municipality owns a plot of land that may be suitable to establish projects, with a hope to expand the municipal boundaries to acquire an additional 100 sq. km of land to establish projects in partnership with the private sector.

1.4.9 Land Use

The Municipality has an updated master plan, legislations and zoning plan, but there is no comprehensive diagram and database to guide investors to suitable plots of land and investment opportunities, whether public or private land.

Agricultural and Forest Area

The agricultural and forest area is very small since it is an industrial area.

Commercial and Industrial Areas

The Municipality of Zarqa is a major industrial area and an active commercial sector. There are 408 large industrial enterprises in the governorate: tanning factories, medical and treatment industries, engineering and IT, plastics, rubber, paper and cardboard. The city also has large commercial areas, as well as 11,000 retail stores, offices and commercial establishments.

Tourist Activities

Zarqa currently does not embody a touristic station.

2 Chapter II: Market Analysis

2.1 Project Description

The idea of the proposed project is to establish a Light Industrial Zone in an area far from the city center, owned by Zarqa Municipality, where it is planned to design the land to be a center for a Light Industrial Zone with an area of 500 donums, with many advantages, including:

- ✓ Attract as many of craftsmen and entrepreneurs as possible in one place.
- ✓ Reduce problems associated with existence of the industrials zone in the city center and mitigate the resulting noise.
- ✓ Possibility of leasing some spaces in the Project for owners of commercial shops, restaurants and cafeterias to revive the area.
- ✓ Contribution of the Project greatly to provide customer service and all kinds of craft services in one place.
- ✓ Activating the hierarchical and interrelating sales in the Project, so that crafts can depend to each other.

The Light Industrial Zone will target craftsmen whose works require availability of different spaces and a number of shoppers and visitors, such as engraving, embossing, blacksmithing, carpentry and others. Location of the Project is characterized by a number of main roads in the Municipality of Zarqa and absence of any similar project in Zarqa or near, in addition to availability of infrastructure required to construct the Project.

2.2 Objectives of the Project

Objectives of the proposed Project are summarized as follows:

- Achieve a profitable return to the related party of the Project.
- Fulfill needs of Zarqa Municipality and adjacent areas of assigned places for industrial crafts in the light of the increasing demand in these areas.
- Activate the industrial sector in Zarqa.
- Provide job opportunities to people of Zarqa.
- Eliminate the industrial craft enterprises from residential communities, due to noise and pollution.

2.3 The Proposed Project

The Project is located on a plot of land, far from the city center of Zarqa, where the land includes spaces with a number of 299 spaces available for lease with a plot of land of 500 donums. 25% of the land area is assigned to cover relegation and rotation area. However, Location of the plot is the most important point of strength of the project, for the following reasons:

- There is no similar project on or near the location within the Municipality jurisdiction and limited availability of commercial stores that are randomly distributed in the Municipality and the province.
- Availability of the infrastructure required.
- Proximity to several main roads in the province.
- Located at the main transport network.

One of the most important justifications for the Project is the Municipality's need to provide new sources of income, which can be utilized to support efforts of the Municipality,

2.4 Targeted Segments

The Project targets entrepreneurs and craftsmen willing to rent commercial spaces in the metropolitan of the town. The Project will also target investor in restaurant business as part of leasing spaces for restaurants, such as cafeterias. The following figure shows the targeted groups through the proposed Project:

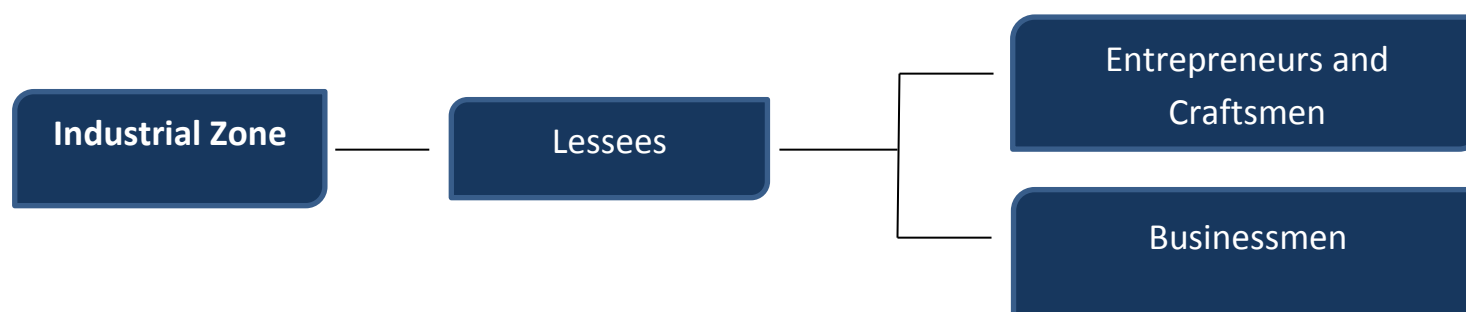


Figure 1: Targeted Segments through the Project

2.5 Market Size Analysis

Craftsmen are very interested to acquire a safe and organized work area to accommodate them and facilitate their tasks. It is very natural for craftsmen to need each other to accomplish a certain work. According to Zarqa Municipality, number of craftsmen in the area is increasing, as number of commercial stores and crafts is increasing, driven by the increase number of population in the area and demand on services and personal commodities, which will lead to an increased competition between craftsmen and entrepreneurs to acquire stores with sufficient paces for their professions.

Table 8: Occupational Licenses Distributed 2014-2016

Seq.	Occupation	2014	2015	2016
1	Brick plants	98	99	111
2	Stones	5	10	3
3	Marble	8	3	6
4	Tiles	24	2	12
Total		135	114	132

2.6 Competitors

Investment in the field of leasing craftsmen stores is one of the suitable investments on plots of land with industrial regulation - industrial zones that contain the following professions:

- Craft workshops (blacksmithing and carpentry).
- Professional works and handcrafts
- Car body straightening and painting
- Vehicles maintenance, equipment and tools.

Competitiveness will be in moving craftsmen and entrepreneurs from their current locations to the proposed Project.

2.7 Estimated Market Share

The market share of the Light Industrial Zone has been built based on survey conducted to collect information about possibility of leasing spaces, within the Light Industrial Zone to craftsmen and entrepreneurs. A sample of 33 craftsmen and owners of commercial stores were surveyed about the project idea and its attractiveness and to possibility obtain a space in the Light Industrial Zone since their current stores/shops are scattered randomly and is causing disturbance to inhabitants and pollution to the environment. As a result of the survey, it was clear that the craftsmen are interested in the idea of the Light Industrial Zone and they are ready to move to it immediately after completion of construction. Based on the Municipality's estimations, number of spaces that will be leased during the first year of operation is 250 spaces. Lessees will be divided into two categories, the first category is craftsmen who are leasing current spaces in the craft areas owned by the Municipality throughout the city (northern and southern); The second category is owners of factories and craftsmen who are leasing plots of land that do not belong to Zarqa Municipality. Based on the data provided by the municipal team, details of the occupied and vacant spaces are as follows:

- Northern Industrial Zone, where the total number of spaces is 125 spaces, divided as follows:
 - a. Occupied spaces: about 105
 - b. Vacant spaces: 30
- Southern Industrial Zone, where the total number of spaces is 66 spaces, divided as follows:
 - a. Occupied spaces: about 64
 - b. Vacant spaces: 2

- The total number of spaces in the northern and southern industrial zone is 191 spaces. Thus, the occupancy rate is 88%.

In addition to the spaces owned and leased by the Municipality at the time being, Zarqa Municipality has a number of craft plants, such as brick plants, and stones saws owned by craftsmen. The Municipality is looking to attract them to the industrial zone to facilitate their businesses, where collection of brick plants can produce a central mixer to serve brick plants, and maintenance workshops for machinery and equipment and commercial enterprises to serve lessees.

2.8 Price Analysis and Estimated Revenue

According to the prevailing rate in the area for the rental price of commercial stores, which is about JD 1,800 per annum, and based on the competitive advantage of the Project, a rental price list was designed in a manner mitigating the competitive advantage. The following table shows a list of rental rates per annum over the ten years of study:

Table 1: Proposed rent rate per store

Year	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Annual rent per space	1,800	1,845	1,891	1,938	1,987	2,037	2,087	2,140	2,193	2,248
Annual rent per space	1,800	1,845	1,891	1,938	1,987	2,037	2,087	2,140	2,193	2,248

Revenue of the Project during the years of study, where rent is subject to annual inflation coefficient of (2.5%) as shown in the following table:

Table 2: Project revenue during years of study

Assumptions of revenue as per occupancy rate	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Number of available spaces	299	299	299	299	299	299	299	299	299	299
Annual rent	1,800	1,845	1,891	1,938	1,987	2,027	2,087	2,140	2,193	2,248
% of occupancy from total available	84%	85%	87%	89%	91%	92%	94%	96%	98%	100%
Total number of spaces	250	255	260	265	271	276	282	287	293	299
Total Revenue of Project in JD	450,000	470,475	491,882	514,262	537,661	562,125	587,701	614,442	642,399	671,628

2.9 SWOT Analysis

Points of Strength	Point of Weakness
• Eliminate the industrial craft enterprises from residential communities, due to noise and pollution. • Contribution of the Project greatly to provide customer service and all kinds of craft services in one place. • Lack of competitiveness in the proposed area. • Providing job opportunities to people of Zarqa Governorate • Prepared sites away from places of overcrowding and traffic congestion, which facilitates works of craftsmen.	• Limited number of stores. • Limited space per store. • Distance is relatively far from customers and targeted groups.
Opportunities	Threats
• Explore possibility of leasing some spaces of the proposed Project for restaurants and cafeterias to revive the area. • Explore possibility of expanding range of the Project to include light industries.	• Emergence of new competitors. • Reluctance of some craftsmen to move from their current locations to the new location.

3 Chapter III - Technical Analysis

3.1 Human Resources Required

The need for workforce is limited to 4 workers. The purpose of their presence is to follow-up the financial matters concerning rents, as well as attracting more lessees and ensure that lessees are committed with their financial and non-financial obligations within their commercial and craft operations in the City. Therefore, workforce required to run the Project is shown in the following organizational structure of the Project:

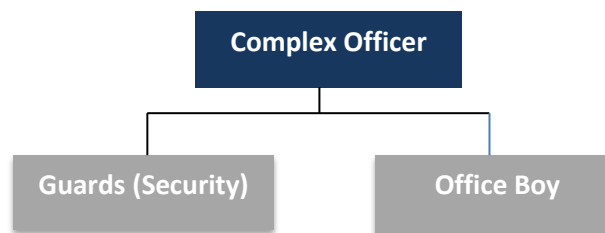


Figure 3 Organizational Structure

The following table shows the direct and indirect employment required for the Project

Table 9: Employment Required for the Project

Job Title	Number
Indirect Employment	
Accountant and Complex Officer	1
Office Boy	1
Guard	5
Total Indirect Employment	7

3.2 Details of Leasable Units (spaces)

The current Light Industrial Zone consists of flat land divided into 299 leasable spaces with a total of 300 donums. 200 donums have been allocated for regulating and sorting. It is expected that 100 other spaces will be sorted within the area allocated for relegations and non-industrial use. These can be utilized later on, such as a clinic, a mosque, a garden, electricity transformers, water and sewage pipelines and electricity. These spaces are expected to be leased for various craft works, especially those planned by Zarqa Municipality to move from population concentrations. In addition to the spaces, an area of 400 m² was allocated as buildings for services and administration offices of the Project.

3.3 Technical Requirements' Analysis

3.3.1 Land and Construction Works

The following table shows the available land area¹⁶:

Table 10: Land required

Land	Cost (JD/m ²)	Area (m ²)	Total Cost/JD
Required plot of land	6	400,000	2,400,000
Total Cost			2,400,000

As for the construction works required for the Project, they are represented by costs of leveling works, paving of spaces' floor, constructing a fence on the plot of land and buildings for services and administration offices. The following table shows the construction costs:

Table 11: Construction works required

Construction Works	Cost (JD/m ²)	Area (m ²)	Total Cost/JD
Leveling works and paving spaces' floor	8	100,000	800,000
A fence surrounding the City estimated by linear meters	40	3,000 (Lm)	120,000
Buildings for services and administration offices	275	800	220,000
Total Cost			1,140,000

¹⁶ Source: Average base price of industrial land in the Municipality of Zarqa, as stated in the Land Department

3.3.2 Furniture and Fixtures

The following table shows the furniture and fixtures required:

Table 12: Furniture and fixtures

Furniture and Fixtures	quantity	Unit Cost (JD)	Total Cost (JD)
Manager desk	1	200	200
Manager chair	3	75	225
Guest chairs	6	50	300
Armchair	5	100	500
Gas cooker and LPG cylinder	3	100	300
Total			1,525

3.3.3 Machinery and Equipment

The following table shows the machinery and equipment required:

Table 13: Machinery and Equipment Required

Furniture and Fixtures	quantity	Unit Cost (JD)	Total Cost (JD)
Air conditioners	5	450	2,250
Total			2,250

3.3.4 Vehicles and Means of Transport

The following table shows the vehicles and means of transport required:

Table 14: Vehicles and Means of Transport

Vehicles and Means of Transport	Quantity	Unit Cost/JD	Total Cost/JD
Medium transport vehicle (Pickup)	1	15,000	15,000
Total Cost			15,000

3.4 Operating Expenses

The following table shows the operating expenses of the Project, where utilities expenses represented the largest proportion of the operating expenses. These represent water and electricity expenses for the shared areas in the proposed Project. They are fully associated with number of users of the lessees, which were estimated at 7%, and maintenance expenses were estimated at 2% of the net book value of the assets as follows:

Table 15: Operating Expenses

Description	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Electricity and water expenses	31,500	32,933	34,432	35,998	37,636	39,349	41,139	43,011	44,968	47,014
Maintenance expenses	0	85,170	87,299	89,482	91,719	94,012	96,362	98,771	101,240	103,771
Direct salaries	33,000	33,759	34,535	35,330	36,142	36,974	37,824	38,694	39,584	40,494
Other expenses	1,290	1,322	1,355	1,389	1,424	1,460	1,496	1,533	1,572	1,611
Total	65,790	153,184	157,622	162,199	166,921	171,794	176,821	182,009	187,364	192,891

3.5 Administrative Expenses

The following table shows the expected administrative expenses of the proposed Project during the years of study. The team of the Professional Group estimated the expenses of the landline phone calls for the Project and bills of cell phones for the employees at a monthly rate of JD 100. As for hospitality and cleaning expenses, they were estimated at JD 150 per month, where the Municipality duties in hygiene matter are limited to waste collection, as follows:

Table 16: Administrative Expenses

Description	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Communications expenses	1,200	1,230	1,261	1,292	1,325	1,358	1,392	1,426	1,462	1,499
Hospitality and cleaning expenses	1,800	1,845	1,891	1,938	1,987	2,037	2,087	2,140	2,193	2,248
Professional charges	2,000	2,050	2,101	2,154	2,208	2,263	2,319	2,377	2,437	2,498
Insurance expenses	115	109	103	97	91	85	78	72	66	60
Total	5,115	5,234	5,356	5,481	5,610	5,742	5,877	6,016	6,158	6,304

3.6 Salaries and Wages

The monthly salary of each employee was estimated to be multiplied by 15 months per year for the purpose of calculating the annual salaries, wages and subsequent social security, health insurance and others. The following tables shows the expected annual salaries and wages, where the rate of increase in wages and salaries per annum is taken into account.

Table 17: Employment required for the Project

Job Title	Number	Salary per Month (JD)	Total Monthly Salaries (JD)	Total Annual Salaries (JD)
Accountant/Complex Officer	1	400	400	6,000
Office Boy	1	300	300	4,500
Guard	5	300	1,500	22,500
Total Salaries	7		2,200	33,000

Table 18: Expected salaries and wages for the ten years

Job Title	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Indirect Employees										
Accountant/Complex Officer	6,000	6,138	6,279	6,424	6,571	6,722	6,877	7,035	7,197	7,363
Office Boy	4,500	4,604	4,709	4,818	4,929	5,042	5,158	5,276	5,398	5,522
Guard	22,500	23,018	23,547	24,088	24,643	25,209	25,789	26,382	26,989	27,610
Total direct salaries	33,000	33,759	34,535	35,330	36,142	36,974	37,824	38,694	39,584	40,494

3.7 Legal Requirements and Procedures

Legal Requirements	Comments
Company Registration	The first step to create a project is establishment of the Project's company through the Ministry of Industry, Trade and Supply according to the required legal form. For the purpose of the feasibility study, establishment of the Project's company was assumed to be a limited liability company.
Construction of Project's buildings	The buildings of the Project (the stores) and other construction works require obtaining construct license from the Municipality, where the process of issuing the construction license requires layouts presented by an engineering office, provided to be verified and stamped by the Jordan Engineers Association.
Issue of occupational license	The occupational license is issued through the Municipality, provided that the requirements of environment, civil defense and the occupational health and safety conditions that observed by the municipalities are achieved. A copy of the commercial registration of the company and the participation receipt of the Chamber of Industry must be accompanied to obtain the vocational license, which qualifies the Project to practice work. The Project is expected to meet the requirements since it will be established under supervision of the Municipality.
Other legal requirements related to objective of the Project	The Project may require additional approvals from the Municipality to operate the Project, such as a central market for vegetables and fruits.

Chapter IV: Financial Analysis

The main object of this study is studying a preliminary economic feasibility for establishment of a Light Industrial Zone in Zarqa Municipality. This study was based on the financial and technical assumptions to obtain the results that help calculating profitability of the Project:

4.1 Assumptions and Facts

The duration of the financial study is 10 years. As for the annual increase rates, the following assumptions were taken as shown on the tables below:

Table 19: Annual increase rates

Annual Increase rates	Annual Rate
Selling prices are increasing annually by ⁽¹⁾	2.50%
Expenses are increasing annually by ⁽²⁾	2.50%
Salaries, wages and equivalents are increasing annually by ⁽³⁾	2.3%

(1) Source: Inflation rate for the last three years - Central Bank

(2) Source: Inflation rate for the last three years - Central Bank

(3) Source: Social Security Corporation

Table 20: Income tax assumptions - salaries - compulsory reserve

Income Tax Assumptions	Amount
Income tax rate in Jordan ⁽¹⁾	14%
Discount percentage of income tax ⁽²⁾	0%
Income tax rate after calculation of discount percentage	14%
Percentage of the compulsory reserve ⁽³⁾	10%

(1) Source: Jordan Income Tax Law No. 34 for the year 2014

(2) Income Tax Reduction Act in the Least Developed Area No. 44 for the year 2016

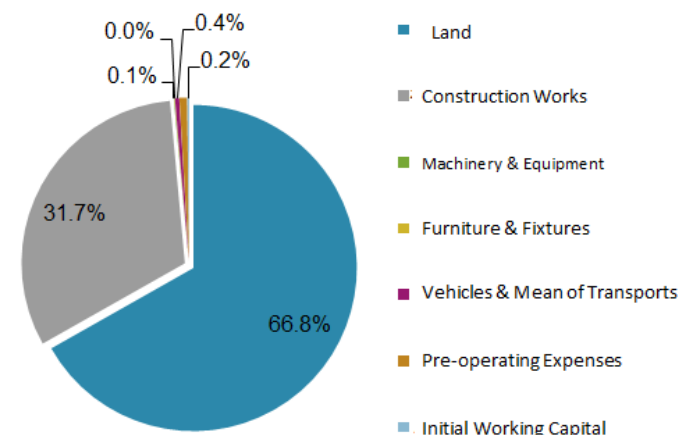
(3) On assumption that the type of company is limited

4.2 Capital Expenditures

The estimated capital cost of the Project is JD 3,774,680, covering cost land, construction works, machinery and equipment, furniture and fixtures, vehicles and means of transport, pre-operating expenses and the initial working capital. Cost of assets is JD 3,736,714, pre-operating expenses are JD 28,875 and the initial working capital is JD 9,092. The following table summarizes the capital expenditures of the proposed Project in Jordanian Dinars:

Table 21: Summary of Capital Expenditures

Summary of capital expenditures	Cost (JD)	Hedging Ratio	Cost (JD)
Land	2,400,000	5%	2,520,000
Construction works	1,140,000	5%	1,197,000
Machinery and equipment	2,250	5%	2,363
Furniture and Fixtures	1,525	5%	1,601
Vehicles and means of transport	15,000	5%	15,750
Pre-operating expenses	27,500	5%	28,875
Initial Working capital	8,659	5%	9,092
Total (JD)	3,594,934		3,774,680



4.2.1 Land

The following table shows the land required for the Project, as follows:

Table 22: Land required for the Project

Land	Cost (JD/m ²)	Area (m ²)	Total Cost (JD)
Plot of land required	6	400,000	2,400,000
Total Cost			2,400,000

4.2.2 Construction Works

The following table shows the construction works required for the Project, which do not need to build pavements, as lease is for the space and the lessee has the right of disposition inside the space allocated for them. The role of the Project is limited in specifying spaces, pathways, roads and services, as follows:

Table 23: Construction works required for the Project

Construct Works	Cost (JD/m ²)	Area (m ²)	Total Cost/JD
Leveling and paving works for pathways' floor	8	100,000	800,000
Fence	40	3,000 (Lm)	120,000
Building for services and administration offices	275	800	220,000
Total Cost			1,140,000

4.2.3 Vehicles and Means of Transport

The following table shows the vehicles and means of transport required for the Project:

Table 24: Vehicles and Means of Transport

Vehicles and Means of Transport	Quantity	Unit Cost/JD	Total Cost/JD
Medium transport vehicle (Pickup)	1	15,000	15,000
Total Cost			15,000

4.2.4 Furniture and Fixtures

The following table shows the furniture and fixtures required for the Project along with the cost:

Table 25: Furniture and fixtures Required for the Project

Furniture and Fixtures	quantity	Unit Cost (JD)	Total Cost (JD)
Manager desk	1	200	200
Manager chair	3	75	225
Guest chairs	6	50	300
Armchair	5	100	500
Gas cooker and LPG cylinder	3	100	300
Total			1,525

4.2.5 Machinery and Equipment

The following table shows the machinery and equipment required:

Table 26: Machinery and Equipment Required

Furniture and Fixtures	quantity	Unit Cost (JD)	Total Cost (JD)
Air conditioners	5	450	2,250
Total			2,250

4.2.6 Pre-operating Expenses

Consist of a set of expenditures that are expensed before commencing the Project and before realizing any revenue. They include the establishing expenses as follows:

Table 27: Pre-operating expenses

Pre-operating Expenses	Estimated Cost (JD)
Government and licenses fees	1,000
Marketing expenses	26,000
Establishing expenses	500
Total	27,500

4.2.7 Initial Working Capital

The initial working capital reflects the Project's need for the required liquidity to cover the costs incurred on operating the Project, until that time when the Project starts collecting money from customers. Since such projects collect revenue on a monthly basis, there is no need for hedging for more than one month. The following table shows components of the initial working capital:

Table 28: Initial working capital

Initial Working Capital	# of Months	Percentage	Estimated Cost (JD)
Operating costs	1	8%	5,483
Administrative expenses	1	8%	426
Direct salaries	1	8%	2,750
Total			8,659

4.3 Depreciation of Assets

The following table summarizes cost of fixed assets, percentage of annual additions and annual depreciation rate:

Table 29: Summary of depreciations and additions expenses on fixed assets

Cost of fittings, capital expenditures and annual depreciation rates	Cost (JD)	Depreciation %	Annual Addition %
Land	2,520,000	0.0%	0.0%
Construction works	1,197,000	5.0%	0.0%
Equipment and Machinery	2,363	10.0%	5.0%
Furniture and fixtures	1,601	10.0%	5.0%
Vehicles and means of transport	15,750	10.0%	0.0%
Total (JD)	3,736,714		

The following tables illustrate the annual increases on fixed assets, depreciations, book value of assets for ten years, from the year of starting operations:

Table 30: Depreciation and annual increase on construction works

Construction Works	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Historical cost	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000
Additions	0	0	0	0	0	0	0	0	0	0
Total	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000	1,197,000
<i>Depreciation %</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>	<i>5%</i>
Annual Depreciation Expenses	59,850	59,850	59,850	59,850	59,850	59,850	59,850	59,850	59,850	59,850
Opening Accumulated Depreciation	0	59,850	119,700	179,550	239,400	299,250	359,100	418,950	478,800	538,650
Ending Accumulated Depreciation	59,850	119,700	179,550	239,400	299,250	359,100	418,950	478,800	538,650	598,500
Net Book Value	1,137,150	1,077,300	1,017,450	957,600	897,750	837,900	778,050	718,200	658,350	598,500

Table 31: Depreciation and annual increase on Machinery and Equipment (JD)

Machinery and Equipment	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Historical cost	2,363	2,363	2,481	2,605	2,735	2,872	3,015	3,166	3,324	3,490
Additions	0	118	124	130	137	144	151	158	166	175
Total	2,363	2,481	2,605	2,735	2,872	3,015	3,166	3,324	3,490	3,665
<i>Depreciation %</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>
Annual Depreciation Expenses	236	248	260	273	287	302	317	332	349	367
Opening Accumulated Depreciation	0	236	484	745	1,018	1,305	1,607	1,924	2,256	2,605
Ending Accumulated Depreciation	236	484	745	1,018	1,305	1,607	1,924	2,256	2,605	2,972
Net Book Value	2,126	1,996	1,860	1,717	1,566	1,408	1,242	1,068	885	693

Table 32: Depreciation and annual increase on Furniture and Fixtures (JD)

Machinery and Equipment	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Historical cost	1,601	1,601	1,681	1,765	1,854	1,946	2,044	2,146	2,253	2,366
Additions	0	80	84	88	93	97	102	107	113	118
Total	1,601	1,681	1,765	1,854	1,946	2,044	2,146	2,253	2,366	2,484
<i>Depreciation %</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>
Annual Depreciation Expenses	160	168	177	185	195	204	215	225	237	248
Opening Accumulated Depreciation	0	160	328	505	690	885	1,089	1,304	1,529	1,766
Ending Accumulated Depreciation	160	328	505	690	885	1,089	1,304	1,529	1,766	2,014
Net Book Value	1,441	1,353	1,261	1,163	1,062	954	842	724	600	470

Table 33: Depreciation and annual increase on Vehicles and Means of Transport (JD)

Machinery and Equipment	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Historical cost	15,750	15,750	15,750	15,750	15,750	15,750	15,750	15,750	15,750	15,750
Additions	0	0	0	0	0	0	0	0	0	0
Total	1,601	1,681	1,765	1,854	1,946	2,044	2,146	2,253	2,366	2,484
<i>Depreciation %</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>	<i>10%</i>
Annual Depreciation Expenses	1,575	1,575	1,575	1,575	1,575	1,575	1,575	1,575	1,575	1,575
Opening Accumulated Depreciation	0	1,575	3,150	4,725	6,300	7,875	9,450	11,025	12,600	14,175
Ending Accumulated Depreciation	1,575	3,150	4,725	6,300	7,875	9,450	11,025	12,600	14,175	15,750
Net Book Value	14,175	12,600	11,025	9,450	7,875	6,300	4,725	3,150	1,575	0

Table 34: Summary of capital costs, additions and annual depreciation (JD)

Summary of capital costs, additions and annual depreciation	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Total Fixed Assets	3,736,714	3,736,912	3,737,120	3,737,339	3,737,568	3,737,809	3,738,062	3,738,327	3,738,606	3,738,899
Total Depreciation Expenses	61,821	61,841	61,862	61,884	61,907	61,931	61,956	61,983	62,011	62,040
Total Accumulated Depreciation	61,821	123,663	185,525	247,408	309,315	371,246	433,202	495,185	557,196	619,236
Total Additions	0	198	208	219	229	241	253	266	279	293
Total Net Book Value	3,674,892	3,613,249	3,551,595	3,489,930	3,428,253	3,366,563	3,304,860	3,243,142	3,181,411	3,119,664

4.4 Financing Resources

The financing structure required for the Project comprises of loans and owners' equity. The investment cost of this Project was estimated at JD 3,774,680. This amount was determined through the fixed assets, which include land, construction works, equipment and machinery, furniture and fixtures, vehicles and means of transport, pre-operating expenses and initial working capital. (30%) of the Project will be financed through loans amounting JD 1,132,404 and the remaining (70%) through self-financing (owner' equity) amounting JD 2,642,276. The following table shows the general structure of funding required:

Table 35: Financing resources required

Financing resources	Amount	Percentage
Owners' equity (self-financing)	2,642,276	70%
Loans	1,132,404	30%
Total	3,774,680	100%
Financing utilization	Amount	Percentage
Capital Expenditures	3,736,714	98.99%
Pre-operating Expenditures	28,875	0.76%
Working Capital	9,092	0.24%
Total	3,774,680	100%

The following table shows the loan assumptions used:

Table 36: Loan assumptions

Loan Assumptions	
Loan Percentage	30%
Owners' equity	70%
Interest rate	6%
Repayment of loan - two years	7
Year of loan	2019
Grace period (one year)	1

The following table shows the schedule financing loan used:

Table 37: Schedule of financing loans used

	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Opening balance	1,132,404	1,132,404	931,520	718,582	492,869	253,612	0	0	0	0
Interests	67,944	55,891	43,115	29,572	15,217	0	0	0	0	0
Capital	0	200,884	212,937	225,714	239,257	253,612	-0	-0	-0	-0
Repayment of loan	67,944	256,776	256,052	255,286	254,473	253,612	0	0	0	0
Ending balance	1,132,404	931,520	718,582	492,869	253,612	0	0	0	0	0
Current part of the loan	200,884	212,937	225,714	239,257	253,612	-0	-0	-0	-0	-0
Long-term loans	931,520	718,582	492,869	253,612	0	0	0	0	0	0

4.5 Annual Revenue

The following table shows the expected annual revenue of the proposed Project, which previously illustrated in the Project's market study, where financial analysis is need only for total revenue:

Table 38: Anticipated Project's revenue for ten years

	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Occupancy rates	84%	85%	87%	89%	91%	92%	94%	96%	98%	100%
Total Revenue of the Project in JD	450,000	470,475	491,882	514,262	537,661	562,125	587,701	614,442	642,399	671,628

4.6 Operating Expenses

The following table shows the expected operating expenses of the proposed Project:

Table 39: Operating Expenses

Description	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Electricity and water expenses	31,500	32,933	34,432	35,998	37,636	39,349	41,139	43,011	44,968	47,014
Maintenance expenses	0	85,170	87,299	89,482	91,719	94,012	96,362	98,771	101,240	103,771
Direct salaries	19,500	19,949	20,407	20,877	21,357	21,848	22,351	22,865	23,391	23,928
Other expenses	1,020	1,046	1,072	1,098	1,126	1,154	1,183	1,212	1,243	1,274
Total	52,050	139,097	143,210	147,455	151,838	156,363	161,035	165,859	170,842	175,988

4.7 Administrative Expenses

The following table shows the expected administrative expenses of the Project during the years of study. They were estimated by the Professional Group's team as follows:

Table 40: Administrative Expenses

Description	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Communications expenses	1,200	1,230	1,261	1,292	1,325	1,358	1,392	1,426	1,462	1,499
Hospitality and cleaning expenses	1,800	1,845	1,891	1,938	1,987	2,037	2,087	2,140	2,193	2,248
Professional charges	2,000	2,050	2,101	2,154	2,208	2,263	2,319	2,377	2,437	2,498
Insurance expenses	115	109	103	97	91	85	78	72	66	60
Total	5,115	5,234	5,356	5,481	5,610	5,742	5,877	6,016	6,158	6,304

4.8 Salaries and Wages

The monthly salary of each employee was estimated to be multiplied by 15 months per year for the purpose of calculating the annual salaries, wages and subsequent social security, health insurance and others. The following tables shows the expected annual salaries and wages, where the rate of increase in wages and salaries per annum is taken into account.

Table 41: Expected salaries and wages for the ten years

Job Title	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Indirect Employees										
Accountant/Complex Officer	6,000	6,138	6,279	6,424	6,571	6,722	6,877	7,035	7,197	7,363
Office Boy	4,500	4,604	4,709	4,818	4,929	5,042	5,158	5,276	5,398	5,522
Guard	22,500	23,018	23,547	24,088	24,643	25,209	25,789	26,382	26,989	27,610
Total direct salaries	33,000	33,759	34,535	35,330	36,142	36,974	37,824	38,694	39,584	40,494

4.9 Statement of Income

Throughout all financial information and analysis of expenses and financial costs of the Project, the estimated statement of income for ten years, shall be as shown in the following table:

Table 42: Statement of Income

Description	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Sales revenue	450,000	470,475	491,882	514,262	537,661	562,125	587,701	614,442	642,399	671,628
Direct operating expenses	(65,790)	(153,184)	(157,622)	(162,199)	(166,921)	(171,794)	(176,821)	(182,009)	(187,364)	(192,891)
Gross profit (or loss)	384,210	317,291	334,260	352,063	370,740	390,331	410,880	432,432	455,035	478,737
<i>Gross profit margin</i>	85%	67%	68%	68%	69%	69%	70%	70%	71%	71%
Administrative and general expenses	(5,115)	(5,234)	(5,356)	(5,481)	(5,610)	(5,742)	(5,877)	(6,016)	(6,158)	(6,304)
Pre-operating expenses	(28,875)									
Total indirect costs	(33,990)	(5,234)	(5,356)	(5,481)	(5,610)	(5,742)	(5,877)	(6,016)	(6,158)	(6,304)
Income before tax, depreciation and interests	350,220	312,056	328,904	346,582	365,130	384,589	405,003	426,417	448,877	472,433
Depreciation of fixed assets	(61,821)	(61,841)	(61,862)	(61,884)	(61,907)	(61,931)	(61,956)	(61,983)	(62,011)	(62,040)
Income before taxes and interests	288,398	250,215	267,042	284,698	303,223	322,659	343,047	364,434	386,866	410,393
Banking interests	(67,944)	(55,891)	(43,115)	(29,572)	(15,217)	(0)	(0)	(0)	(0)	(0)
Income before taxes	220,454	194,324	223,927	255,126	288,006	322,659	343,047	364,434	386,866	410,393
Income tax	(30,864)	(27,205)	(31,350)	(35,718)	(40,321)	(45,172)	(48,027)	(51,021)	(54,161)	(57,455)
Net profit (loss)	189,590	167,118	192,577	219,408	247,686	277,486	295,021	313,413	332,705	352,938
<i>Net profit rate</i>	42%	36%	39%	43%	46%	49%	50%	51%	52%	53%
Compulsory reserve	(18,959)	(16,712)	(19,258)	(21,941)	(24,769)	(27,749)	(29,502)	(31,341)	(33,270)	(35,294)
Retained earnings	170,631	321,038	494,357	691,825	914,742	1,164,479	1,429,998	1,712,070	2,011,504	2,329,149

4.10 Statement of Financial Position

The statement of financial position is one of the essential financial statements, on which the Project relies on. It is a statement or a report illustrates the financial position of the Project. The statements are prepared based on the assumptions of the accounts receivable and payable of the Project.

4.10.1 Accounts Receivable and Payable

The following table shows the assumptions of the accounts receivable and payable and ending inventory:

Table 43: Assumptions of accounts receivable and payable

Assumption of Balance Sheet	Percentage
Cash collected during the period from the accrued revenue	75%
Cash collect during the subsequent period	25%
Cash paid on the operating expenses during the fiscal year	100%
Cash paid during the subsequent period	0%

Table 44: Accounts receivable for the ten years

Accounts receivable	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Operating revenue	450,000	470,475	491,882	514,262	537,661	562,125	587,701	614,442	642,399	671,628
Cash collected during the period	337,500	352,856	368,911	385,697	403,246	421,594	440,776	460,831	481,799	503,721
Cash collected from the previous period	0	112,500	117,619	122,970	128,566	134,415	140,531	146,925	153,610	160,600
Cash collected	337,500	465,356	486,530	508,667	531,811	556,009	581,307	607,757	635,410	664,321
Opening accounts receivable	0	112,500	117,619	122,970	128,566	134,415	140,531	146,925	153,610	160,600
Ending accounts receivable	112,500	117,619	122,970	128,566	134,415	140,531	146,925	153,610	160,600	167,907

Table 45: Movement and balances of accounts payable for the ten years

Accounts Payable	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Operating expenses	32,520	119,149	122,803	126,578	130,481	134,514	138,684	142,994	147,451	152,059
Cash paid during the period	32,520	119,149	122,803	126,578	130,481	134,514	138,684	142,994	147,451	152,059
Cash paid from the previous period	0	0	0	0	0	0	0	0	0	0
Cash Paid	32,520	119,149	122,803	126,578	130,481	134,514	138,684	142,994	147,451	152,059
Opening accounts payable	0	0	0	0	0	0	0	0	0	0
Ending accounts payable	0	0	0	0	0	0	0	0	0	0

The following table shows the statement of financial position of the Project during the years of study in JD:

Table 46: Statement of Financial Position

Description	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Assets										
Current Assets										
Cash	176,878	199,637	235,579	285,343	349,600	429,048	779,378	1,147,823	1,535,271	1,942,649
Accounts receivable	112,500	117,619	122,970	128,566	134,415	140,531	146,925	153,610	160,600	167,907
Total Current Assets	289,378	317,256	358,549	413,909	484,015	569,580	926,303	1,301,434	1,695,870	2,110,556
Non-current Assets										
Fixed Assets (at Net)	3,674,892	3,613,249	3,551,595	3,489,930	3,428,253	3,366,563	3,304,860	3,243,142	3,181,411	3,119,664
Total Non-current Assets	3,674,892	3,613,249	3,551,595	3,489,930	3,428,253	3,366,563	3,304,860	3,243,142	3,181,411	3,119,664
Total Assets	3,964,271	3,930,505	3,910,144	3,903,839	3,912,268	3,936,142	4,231,163	4,544,576	4,877,281	5,230,219
Liabilities										
Current Liabilities										
Accounts payable	0	0	0	0	0	0	0	0	0	0
Current part of the loan	200,884	212,937	225,714	239,257	253,612	(0)	(0)	(0)	(0)	0
Total Current Liabilities	200,884	212,937	225,714	239,257	253,612	(0)	(0)	(0)	(0)	0
Non-current Liabilities										
Long-term Loans	931,520	718,582	492,869	253,612	0	0	0	0	0	0
Total long-term liabilities	931,520	718,582	492,869	253,612	0	0	0	0	0	0
Total Liabilities	1,132,404	931,520	718,582	492,869	253,612	0	0	0	0	0
Owners' Equity										
Shares of members	2,642,276	2,642,276	2,642,276	2,642,276	2,642,276	2,642,276	2,642,276	2,642,276	2,642,276	2,642,276
Compulsory reserve	18,959	35,671	54,929	76,869	101,638	129,387	158,889	190,230	223,500	258,794
Retained profit (loss)	170,631	321,038	494,357	691,825	914,742	1,164,479	1,429,998	1,712,070	2,011,504	2,329,149
Total Owners' Equity	2,831,867	2,998,985	3,191,562	3,410,970	3,658,656	3,936,142	4,231,163	4,544,576	4,877,281	5,230,219
Total Liabilities and Owners' Equity	3,964,271	3,930,505	3,910,144	3,903,839	3,912,268	3,936,142	4,231,163	4,544,576	4,877,281	5,230,319

4.10.2 Statement of Expected Cash Flows

The following table shows the Project's statement of expected cash flows. It indicates that cash flows over all the study period are positive:

Table 47: Statement of Expected Cash Flows

Currency (JD)	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Cash flows from operating activities										
Net profit (loss)	189,590	167,118	192,577	219,408	247,686	277,486	295,021	313,413	332,705	352,938
Non-cash adjustments										
Bank interests	67,944	55,891	43,115	29,572	15,217	0	0	0	0	0
Depreciations	61,821	61,841	61,862	61,884	61,907	61,931	61,956	61,983	62,011	62,040
Operating cash flows before changes in working capital	319,356	284,851	297,554	310,864	324,809	339,417	356,977	375,396	394,716	414,978
Shortage (Excess) in accounts receivable	(112,500)	(5,119)	(5,352)	(5,595)	(5,850)	(6,116)	(6,394)	(6,685)	(6,989)	(7,307)
Shortage (Excess) in accounts payable	0	0	0	0	0	0	0	0	0	0
Changes in working capital	(112,500)	(5,119)	(5,352)	(5,595)	(5,850)	(6,116)	(6,394)	(6,685)	(6,989)	(7,307)
Net cash flows (used) resulted from operating activities	206,856	279,732	292,202	305,269	318,959	333,301	35,583	368,711	387,726	407,671
Cash flows from investment activities										
(Purchase) fixed assets	(3,736,714)	(198)	(208)	(219)	(229)	(241)	(253)	(266)	(279)	(293)
Net cash flows (used) resulted from investment activities	(3,736,714)	(198)	(208)	(219)	(229)	(241)	(253)	(266)	(279)	(293)
Cash flows from financing activities										
Capital	2,642,276									
Repayment of loan	0	(200,884)	(212,937)	(225,714)	(239,257)	(253,612)	0	0	0	0
Bank interests	(67,944)	(55,891)	(43,115)	(29,572)	(15,217)	(0)	(0)	(0)	(0)	(0)
Loans	1,132,404	0	0	0	0	0	0	0	0	0
Net cash flows (used) resulted from financing activities	3,706,736	(256,776)	(256,052)	(255,286)	(254,473)	(253,612)	0	0	0	0
Net excess (shortage) in cash	176,878	22,758	35,942	49,765	64,257	79,449	350,330	368,445	387,447	407,378
Opening cash flows	0	176,878	199,637	235,579	285,343	349,600	429,048	779,378	1,147,823	1,535,271
Ending cash flows	176,878	199,637	235,579	285,343	349,600	429,048	779,378	1,147,823	1,535,271	1,942,649

4.11 Financial Ratios and Profitability

In order to reach to the statement of discounted cash flows, the discount coefficient must be calculated, where there is more than one method to calculate it, but in this study the Weighted Average Cost of Capital (WACC) was used as a discount coefficient to calculate the discounted cash flows and net current value of investment.

The company's financing structure relies on owner's equity and loans. Therefore, the discount rate should be adjusted according to WACC. This is based on the following assumptions:

Table 48: Assumptions of Financial Ratios and Profitability

Risk Allowance	
Revenue rate on risk-free investments (1)	5.80%
Revenue to maturity of debts (2)	9.48%
Market risk allowance (3)	13.20%
Income tax rate	14.00%
Beta "Systemic Risk (4)	0.7
Growth rate	3.00%

(1) Source: Central Bank of Jordan

(2) Source: Central Bank of Jordan

(3) Source: Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source: Damodoran Beta By Sector

According, the WACC of the Project is 12.97%, as shown in the following table:

Table 49: Weighted Average Cost of Capital (WACC)

Weighted Average Cost of Capital	
Revenue rate on risk-free investments	5.80%
Revenue to maturity of debts	9.48%
Market risk allowance	13.20%
Income tax rate	14%
Beta "Systemic Risk	0.70
Owners' equity	2,640,606
Value of loans	1,131,688
Loans	
Cost of borrowing before taxes	9.48%
Cost of borrowing after taxes	8.15%
Value of loan	1,131,688
Percentage of loan	30%
Owner's equity	
Cost of owner' equity	15.0%
Value of owners' equity	2,640,606
Percentage of owners' equity	70%
Total	
Project's value	3,772,294
Project's percentage	100%
Weighted Average Cost of Capital	12.97%

Based on the above, the following table shows the net expected free cash flows statement of the Project:

Table 50: Statement of Expected Net Free Cash Flows of the Project (Return on Investment)

Return on Investment (Currency JD)	0	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year	
Income before taxes and interests		317,272	250,215	267,042	284,698	303,223	322,659	343,047	364,434	386,866	410,393	
Income tax		(30,864)	(27,205)	(31,350)	(35,718)	(40,321)	(45,172)	(48,027)	(51,021)	(54,161)	(57,455)	
Net Profit		286,410	223,010	235,692	248,980	262,902	277,486	295,021	313,413	332,705	352,938	Remaining value
Adjustments												
Depreciation		61,821	61,841	61,862	61,884	61,907	61,931	61,956	61,983	62,011	62,040	
Changes in working capital		(112,500)	(5,119)	(5,352)	(5,595)	(5,850)	(6,116)	(6,394)	(6,685)	(6,989)	(7,307)	
Additions of fixed assets and capital		0	(198)	(208)	(219)	(229)	(241)	(253)	(266)	(279)	(293)	
Net free cash flows	(3,774,680)	235,731	279,534	291,994	305,051	318,730	333,060	350,330	368,445	387,447	407,378	4,206,998

Table 51: Net Cash Flows of the Project

Net Cash Flows	0	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year	Remaining value
Net free cash flows	(3,774,680)	235,731	279,534	291,994	305,051	318,730	333,060	350,330	368,445	387,447	407,378	4,206,998
Discount coefficient	1.00	0.89	0.78	0.69	0.61	0.54	0.48	0.43	0.38	0.33	0.30	0.30
Net current value of cash flows	(3,774,680)	208,660	219,017	202,507	187,267	173,194	160,198	149,153	138,851	129,244	120,287	1,242,205

The following table shows the net expected free cash flows - owners' equity:

Table 52: Statement of Net Free Cash Flows of the - Owners' Equity

Currency JD	0	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year	
Income before tax		220,454	194,324	223,927	255,126	288,006	322,659	343,047	364,434	386,866	410,393	
Income tax		(30,864)	(27,205)	(31,350)	(35,718)	(40,321)	(45,172)	(48,027)	(51,021)	(54,161)	(57,455)	
Net Profit		189,590	167,118	192,577	219,408	247,686	277,486	295,021	313,413	332,705	352,938	Remaining value
Adjustments												
Depreciation		61,821	61,841	61,862	61,884	61,907	61,931	61,956	61,983	62,011	62,040	
Changes in working capital		(112,500)	(5,119)	(5,352)	(5,595)	(5,850)	(6,116)	(6,394)	(6,685)	(6,989)	(7,307)	
Settlement of loan		0	(200,884)	(212,937)	(225,714)	(239,257)	(253,612)	0	0	0	0	
Additions of fixed assets and capital		0										
Net free cash flows	(2,642,276)	138,912	23,155	36,358	50,202	64,715	79,930	350,836	368,976	388,005	407,964	4,213,046

Table 53: Net Cash Flows - Owners' Equity

Net Cash Flows	0	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year	Remaining value
Net free cash flows	(2,642,276)	138,912	23,155	36,358	50,202	64,715	79,930	350,836	368,976	388,005	407,964	4,213,046
Discount coefficient	1.00	0.89	0.78	0.69	0.61	0.54	0.48	0.43	0.38	0.33	0.30	0.30
Net current value of cash flows	(2,642,276)	122,959	18,142	25,215	30,818	35,166	38,445	149,368	139,051	129,430	120,460	1,243,991

The results of expected financial ratios' analysis and profitability of the Project can be seen in the following table:

Table 54: Expected financial ratios and profitability

Expected financial ratios and profitability	On Investment	On Owners' Equity
Discount rate	12.97%	12.97%
Net current value of cash flows	(844,097)	(589,229)
Capital recovery period	More than 10 years	More than 10 years
Internal rate of return (IRR)	9.11%	9.75%

Based on the financial analysis results, it was concluded that the Project is profitable and economically infeasible, due to the high volume of investment.

Occupancy rates Scenario: Assuming a change in project occupancy rates, revenue is as shown in the following table:

Table 55: Occupancy Rates Scenario

Revenue Assumptions	1 st year	2 nd year	3 rd year	4 th year	5 th year	6 th year	7 th year	8 th year	9 th year	10 th year
Occupancy rate of total available	25%	50%	75%	90%	100%	100%	100%	100%	100%	100%
Total revenue of the Project in JD	134,550	275,828	424,085	521,624	594,072	608,924	624,147	639,751	655,744	672,138

Accordingly, the expected financial ratios and profitability are as follows:

Table 56: Expected Financial Ratios and Profitability for Occupancy Rates Scenario

Expected Financial Ratios and Profitability	Investment	Owners' Equity
Discount Rate	12.97%	12.97%
Net current Cash Flows Value	(1,131,024)	(876,301)
Duration of Capital Recovery	More than 10 years	More than 10 years
IRR	8.05%	8.48%

Supplement 1: Questionnaire Form of the Municipality of Zarqa on Northern and Southern Brick Plants

Zarqa Municipality

Environment Department

Questionnaire

1. Name of Investor:
2. Contractual Value:
3. Occupation:
4. Project Expenses of Electricity (monthly in kilowatts, JD)
.....
.....
5. Project Expenses of Water (monthly in liter, JD)
.....
.....
6. Current status of services available (roads, water, electricity, etc.)
.....
.....
7. Satisfaction of the neighboring inhabitants of the current project

.....

.....

8. Plot Number Area.....

.....

.....

a. Is the space sufficient?

.....

.....

b. If a new location is secured, what are the spaces required for your project?

.....

.....

c. What are the proposed locations to establish a specialized Light Industrial Zone?

.....

.....

d. What are the proposed specifications for the new location?

.....

.....

9. What are the safety and security precautions available at the existing location?

.....

.....

10. Any other suggestions?

.....

.....

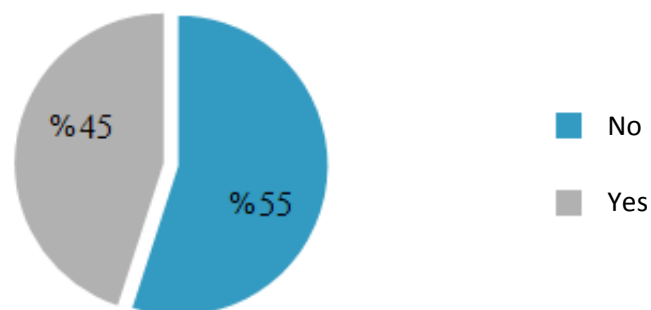
Thank you for your kind cooperation

Supplement 2: Illustrations on the most important results of Zarqa Municipality Questionnaire

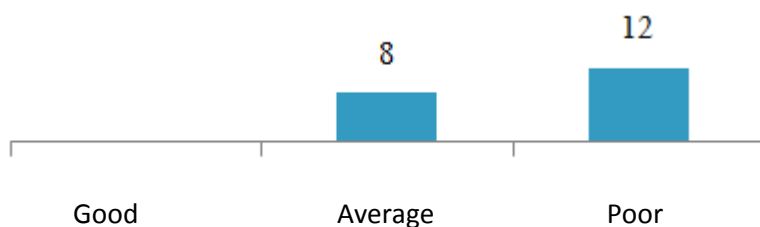
Northern brick plant

The information was collected from a sample of 20 lessees to inquire about the current location and measure the idea of the new location. The following drawings illustrate the results of the questionnaire analysis:

Is the current space sufficient?



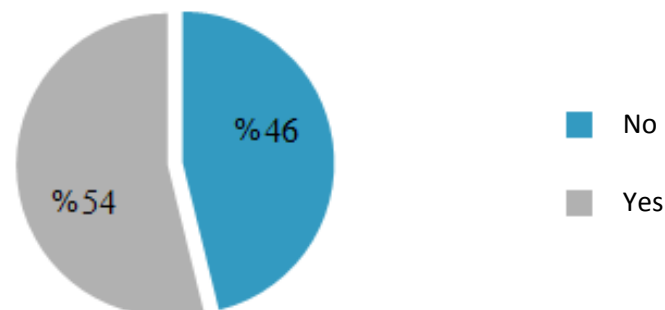
Quality of service in the current location



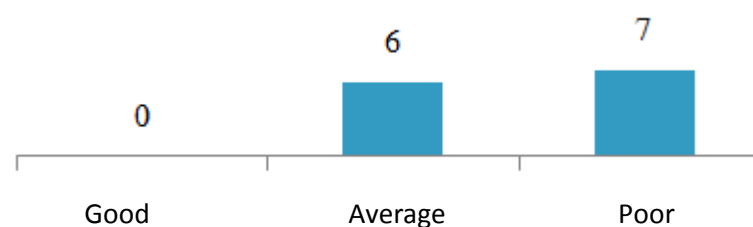
Southern Brick Plant

The information was collected from a sample of 13 lessees to inquire about the current location and measure the idea of the new location. The following drawings illustrate the results of the questionnaire analysis:

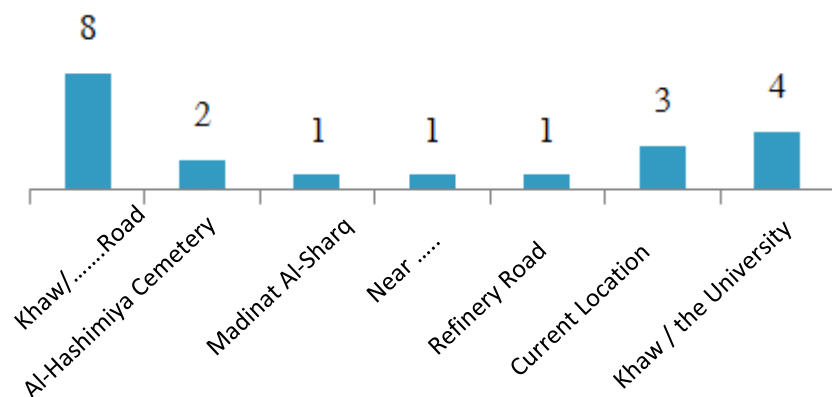
Is the current space sufficient?



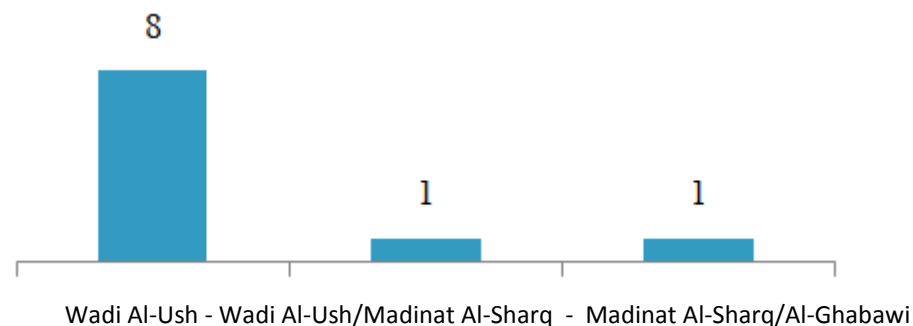
Quality of service in the current location



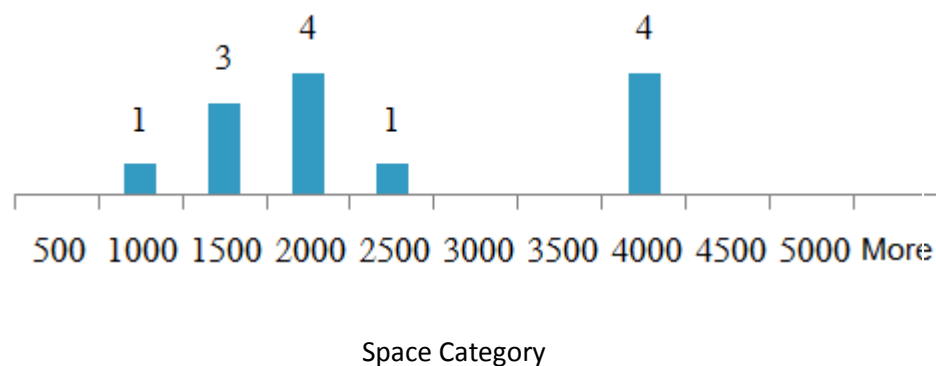
Proposed Locations for the Project



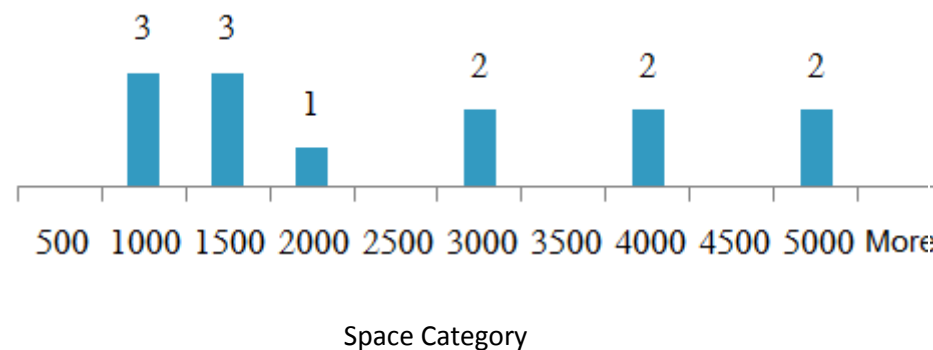
Proposed Locations for the Project



Required Spaces



Required Spaces



Notice

This Preliminary Feasibility Study report (the "Report" or "Study" was presented to the Local Development Division in Zarqa Municipality. The sole objective of this study is to assist the Municipality in assessing the financial feasibility and profitability of this project and demonstrate the various aspects and economic results, associated with the Project.

The anticipated financial information included in this report is based on a number of assumptions and estimations, which although considered logical by the financial advisor, they are fundamentally vulnerable to business, economic and competitiveness uncertainties, all of which are difficult to predict and many beyond the control of the study authors. Accordingly, there can be no guarantee that the expected results will be realized, or that actual results will not be significantly greater or lower than expectations. As for the accuracy or completeness of the information contained in this report (including the expected results or estimates and assumptions on which it is built), the authors of study expressly deny any part or all of the liability that maybe based on such information, errors or omissions about it.

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